

OOP and Design Patterns (CSCI 375)
Student Showcase (Final Project) Rubric

1. Project Title: File Sorter Application

Instructions:

1. There are 10 technical requirements (worth 100 points) to grade the project and the team presentation.
2. For each requirement, use 0 - 10 scale in the Score column (0-6: Fail, 7: C (Average), 8: B (Above Average), 9: A (Good), (10: Excellent)
3. Use the *Notes* section to jot down any observations that may help in grading and justification.

Team and Technical Project Requirement	Score
1. Use 4+1 Views to explain the software design to the audience. Notes: While we didn't show all of the various views, we picked 2 of the 4 we had prepared, in the interest of time.	10/10
2. Use of 3 Design Patterns -- presentation clearly stated and briefly explained design patterns use. Common design patterns are Iterator, Decorator, Observer, Strategy, Command, State, Singleton, Adapter, Façade, Flyweight, Abstract Factory, Composite, Template, MVC, etc. Notes: We used 3 types of design patterns, I made a singleton class, while the other group members made other ones	9/10
3. Use of Fundamental OOD Concepts- e.g., Inheritance, Abstraction, Getters and Setters, Overloading, Attributes and Methods, etc. Notes: We used getters and setters, and attributes and methods	9/10
4. Software management – good usage of management, communication and tracking tools e.g., Gant chart, Kanban board, GitHub Project, Clickup, Discord, Slack, etc. Notes: We used Github to efficiently to track everyone's work and use it to work on various issues. We talked through text and meet-ups in person to communicate ideas and what we worked on	10/10
5. Documentation – clear, easy to follow documentation, UML diagrams are complete, and notations are correct; explanation of objects interaction is clear and complete.	10/10

Notes: We provide documentation and UML diagrams to show how the project functions	
6. Testing and CI-CD pipeline – automatically generates test data using hypothesis, usage of mocking/patching, provides code coverage and Python type check (mypy) reports, CI-CD, Docker, etc. Notes: We implemented testing and have a 100% coverage score	10/10
7. Clean code and Modularity – project uses appropriate software and system design; the code follows best practices (pep8), is well organized and refactored into packages, modules, classes, etc. Notes: The code could probably stand to be refactored, but it's all designed to work well when it does. We also used flake8, and mypy to design the code	9/10
8. Project requirements and execution -- clearly stated functional and technical requirements, project adequately challenging for sophomore-junior students; project demo was clear and concise, etc. Notes: We clearly followed the project requirements and the code performs well	10/10
9. Team presentation -- all members participated in presentation, used the visual and oral presentation techniques and tools to engage audience, etc. Notes: All team members presented and described various aspects of the code and project functionality	10/10
10. Individual Contributions – briefly explain how and what each member contributed to the successful execution of the project. Notes: I worked on the Front-End of the project making a GUI for users to interact with the program in a clean way, I didn't use AI in any fashion to work on my part of the project, however my group members did use AI to work on the backend. Brian worked on most of the backend functionality designing the save functions and making files for the tagging system. Parker assisted him with any issues he had.	10/10
11. BONUS: Above and beyond – Team went beyond the above list e.g., great User Interface, use of Database, security, real-world application, real-world client delight and interaction, etc.	5/10

Notes: I think this does have a real world application and it could certainly be fashioned to be use a Database, Which I did encourage, but Brian wanted to challenge himself by designing his own kind of database	
Total Score Note: Max score can be 110 due to 10 BONUS points.	102/100